

Thermo HW 3 95.4%

Andrey
Stump

① Determine specified properties using tables.

a) Water @ $P=100$ lbf/in² & $u=500$ Btu/lb, Find T in °F, v in ft³/lb & h in Btu/lb

$$T = 327.9^\circ\text{F} \quad X = \frac{u - u_f}{u_g - u_f} = \frac{500 - 298.3}{1105.8 - 298.3} = 0.25 \quad 0.25 = \frac{v - 0.01774}{0.434 - 0.01774} \quad v = 1.12 \text{ ft}^3/\text{lb}$$

$$0.25 = \frac{h - 298.6}{1187.8 - 298.6} \quad h = 520.9 \text{ Btu/lb}$$

b) ammonia @ $T=40^\circ\text{F}$ & $x=0.5$. Find P in lbf/in², v in ft³/lb & u in Btu/lb

$$P = 73.3 \text{ lbf/in}^2 \quad v = 1.9976 \text{ ft}^3/\text{lb} \quad u = 327.69 \text{ Btu/lb}$$

c) Water @ $P=8$ bar & $v=0.0434$ m³/kg. Find T in °C, u in kJ/kg, & h in kJ/kg

$$T = 480^\circ\text{C} \quad u = 3030 \text{ kJ/kg} \quad h = 3350 \text{ kJ/kg}$$

d) Refrigerant 134a at $T=-20^\circ\text{C}$ & $h=235.31$ kJ/kg. Find P in bar, v in m³/kg & u in kJ/kg

$$P = 1.33 \text{ bar} \quad v = 0.145 \text{ m}^3/\text{kg} \quad u = 216 \text{ kJ/kg}$$

② Water contained in a closed, rigid tank, initially SV at 180°C , is cooled to 100°C . Find a) P_0 in MPa b) final quality

State 1	State 2	$X = \frac{v - v_f}{v_g - v_f} = \frac{0.1941 - 1.0435 \times 10^{-3}}{1.673 - 1.0435 \times 10^{-3}} = 0.1195$
$N = 0.1941 \text{ m}^3/\text{kg}$	$N = 0.1941 \text{ m}^3/\text{kg}$	
$T_0 = 180^\circ\text{C}$	$T_f = 100^\circ\text{C}$	$P_i = 10.02 \text{ bar} = 1.002 \text{ MPa}$

③ 2 kg of Water-vapor in piston-cylinder assembly expands at a constant P of 300 kPa (3.0 Bar) from SV state to V of 2.064 m^3 . Find $T_0(^\circ\text{C})$, $T_f(^\circ\text{C})$, W (kJ).

State 1	State 2	$\frac{P_1 v_1}{T_1} = \frac{P_2 v_2}{T_2}$	$W = -P(v_2 - v_1)$
$T_1 = 133.6^\circ\text{C}$	$V = 2.064 \text{ m}^3$	$T_2 = 227.5^\circ\text{C}$	$W = -300(2.064 - 1.2116) = -255.72 \text{ kJ}$
$\bar{v} = 0.6058 \text{ m}^3/\text{kg}$	$P = 300 \text{ kPa}$		
$V = 1.2116 \text{ m}^3$			

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④ Refrigerant 134a expands in a piston-cylinder assembly $P_1=10$ bar & $T_1=60^\circ\text{C}$ to $P_2=2$ bar & $T_2=30^\circ\text{C}$. Mass of refrigerant is 1 kg. During, work done by refrigerant is 20 kJ. KE & PE are negligible. Find a) V_1 in m³ b) Q in kJ

$P_1=10 \text{ bar}$	$P_2=2 \text{ bar}$	$Q+W = \Delta U + \Delta KE + \Delta PE$	$X_1 = \frac{2.77 \times 10^{-3} - 0.945 \times 10^{-3}}{0.0114 - 0.945 \times 10^{-3}} = 0.714$
$T_1=60^\circ\text{C}$	$T_2=30^\circ\text{C}$	$Q + 20 = (159.37 - 156.87)$	$X_2 = \frac{0.0116 - 0.8417 \times 10^{-3}}{0.0165 - 0.8417 \times 10^{-3}} = 0.45$
$R=8.314$		$Q = 22.487 \text{ kJ}$	$0.45 = \frac{u - 139.82}{156.81 - 139.82} \quad u_1 = 156.87 \text{ kJ/kg}$
			$0.45 = \frac{u_f - 96.81}{159.357 - 96.81} \quad u_2 = 159.357 \text{ kJ/kg}$
$(10 \times 10^5)(V_1) = 1(8.314)(333.15)$			
$V_1 = 2.77 \times 10^{-3} \text{ m}^3$			

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- 5) closed, rigid tank contains 1.5 kg of water, initially a 2-phase LV mixture @ $T_1 = 70^\circ\text{C}$. Heat transfer occurs until tank contains only SV at $T_2 = 120^\circ\text{C}$. Find a) P_1 in kPa b) Q in kJ

State 1 MW State 2 JV $x = \frac{0.8919 - 1.028 \times 10^{-5}}{5.042 - 1.028 \times 10^{-5}} = 0.1767$

$T_1 = 70^\circ\text{C}$ $T_2 = 120^\circ\text{C}$ $0.1767 = \frac{u_1 - 292.95}{2464.6 - 292.95} \rightarrow u_1 = 677.51 \text{ kJ/kg}$ $Q = 1851.34 \times 1.5$

$v = 0.8919 \text{ m}^3/\text{kg}$ $v = v_g = 0.8919 \text{ m}^3/\text{kg}$ $+W - \Delta U + \Delta KE + \Delta PE$ $Q = u \cdot m$

$P_1 = 31.19 \text{ kPa}$ $u_2 = 2528.9 \text{ kJ/kg}$ $q = 2528.9 - 677.51 = 1851.34 \text{ kJ/kg} = q$ $Q = 2777.01 \text{ kJ}$

- 6) tank contains 2-phase LV mixture of refrigerant 22 at 10 bar. Mass of SL in tank is 25 kg & quality is 60%. Find a) V in m^3 b) Frac of V_{total} occupied by SV

$P_1 = 10 \text{ bar}$ $V = 0.8352 \times 10^{-3} + 0.6(0.0236 - 0.8352 \times 10^{-3})$ $V_g = 0.0236 \times 37.5 = 0.885 \text{ m}^3$

$v_f = 0.8352 \times 10^{-3} \text{ m}^3/\text{kg}$ $V = 0.01449 \text{ m}^3/\text{kg}$ $v_g = 0.8352 \times 10^{-3} \times 25 = 0.02088 \text{ m}^3$

$v_g = 0.0236 \text{ m}^3/\text{kg}$ $x = \frac{m_g}{m_g + m_f} \rightarrow 0.6 = \frac{m_g}{m_g + 25}$ $V = m \cdot v = (25 + 37.5)(0.01449) = 0.9056 \text{ m}^3$

$V = v_f + x(v_g - v_f)$ $m_g = 37.5 \text{ kg}$ $x_{\text{frac}} = \frac{v_g}{V} = \frac{0.885}{0.9056} \times 100 = 97.725\%$

- 7) balloon filled w/ 0.2 grams of Helium, $T_1 = 27^\circ\text{C}$, 1.3 bar, is released & rises until helium is $T_2 = 17^\circ\text{C}$, $P_2 = 0.8 \text{ bar}$. Assume helium behaves as ideal gas. Find a) R b) V_1 c) V_2

$P_1 = 1.3 \text{ bar} = 130 \text{ kPa}$ $P_2 = 0.8 \text{ bar} = 80 \text{ kPa}$ $R = \frac{8.314}{4.003} = 2.077 \text{ kJ/kg} \cdot \text{K} = 2.077 \text{ J/g} \cdot \text{K}$

$T_1 = 27^\circ\text{C} = 300 \text{ K}$ $T_2 = 17^\circ\text{C} = 290 \text{ K}$ $P_1 V_1 = m R T_1 \rightarrow 130 V_1 = 2 \times 10^{-4} (2.077)(300)$ $V_1 = 958.6 \text{ cm}^3$

$m = 0.2 \text{ g} = 2 \times 10^{-4} \text{ kg}$ $P_2 V_2 = m R T_2 \rightarrow 80 V_2 = 2 \times 10^{-4} (2.077)(290)$ $V_2 = 1505.8 \text{ cm}^3$

- 8) CO_2 fills a closed, rigid tank fitted w/ paddle wheel, $T_1 = 25^\circ\text{C}$, $P_1 = 2.75 \text{ bar}$, $V_1 = 0.05 \text{ m}^3$. Gas is stirred until $T_2 = 257^\circ\text{C}$. During process $Q = 2.75 \text{ kJ}$. Ideal gas, not constant spec heats. KE & PE ignored. Find a) m_{CO_2} in kg b) W in kJ

$T_1 = 25^\circ\text{C} = 298.15 \text{ K}$ $R_{\text{CO}_2} = \frac{8.314}{44.01} = 0.1889 \text{ kJ/kg} \cdot \text{K}$

$P_1 = 2.75 \text{ bar} = 275 \text{ kPa}$ $P_1 V_1 = m R_{\text{CO}_2} T_1 \rightarrow 275(0.05) = m(0.1889)(298.15)$

$V_1 = 0.05 \text{ m}^3$ $m_{\text{CO}_2} = 0.2411 \text{ kg}$

$T_2 = 257^\circ\text{C} = 530.15 \text{ K}$ $Q = 2.75 \text{ kJ}$

$V_2 = 0.05 \text{ m}^3$ $\Delta Q = \Delta U + W$

$-2.75 = m c_v (T_2 - T_1) + W$

$-2.75 = (0.2411)(0.05)(530.15 - 298.15) + W$

$W = -5.582 \text{ kJ}$

80%